Investigation A; Wheel Odometry

Test A1; Linear Displacement Accuracy

Design a test node that drives the robot straight forward for a fixed speed and duration corresponding to a target distance.
Two distances were selected to capture both short-range accuracy and long-distance error accumulation. Additional intermediate distances were not included, as the selected values were sufficient to demonstrate the trend in odometry performance while allowing more trials per condition to improve statistical reliability.

Pass/Fail criterion: % error < 5 %
Forward Speed: 0.2 m/s

Trial Number	Trial Target (m)	Odom reported (m)	Physical measurement (m)	% Error	Pass / Fail
1	5m	4.9719	5.01	0.76	Pass
2	5m	4.9926	5.03	0.74	Pass
3	5m	4.9914	5.01	0.37	Pass
4	5m	4.9926	5	0.15	Pass
5	5m	5.0118	5.02	0.16	Pass
6	1m	1.0178	1.03	1.18	Pass
7	1m	0.9471	1.01	6.23	Fail
8	1m	0.9769	1	2.31	Pass
9	1m	1.006	1.01	0.40	Pass
10	1m	0.9888	1	1.12	Pass

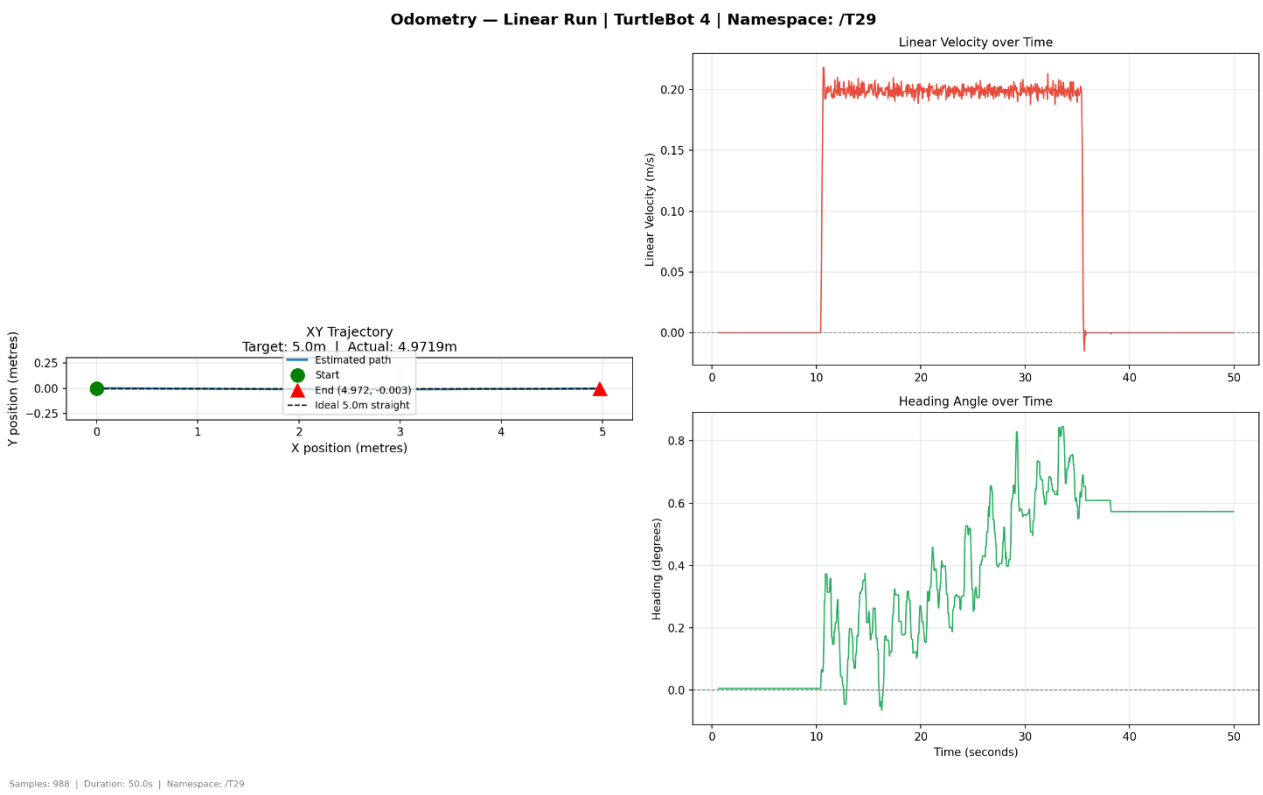
Metric	Formula	What it tells you
Mean ($\bar{x}$)	$\bar{x} = (x_1 + x_2 + \ldots + x_n) / n$	The average result; how close the sensor is to truth on average
Standard Deviation (σ)	$\sigma = \text{sqrt}(\Sigma(x_i - \bar{x})^2 / n)$	How consistent the readings are across trials; sensor precision
Error	error = $\bar{x}$ − ground truth	Signed difference between average reading and known true value
% Error	% error = |error| / ground truth × 100	Normalised accuracy; allows comparison across different distances

Summary statistics (5m):

Metric	Value
Mean displacement (m):	**4.992**
Std dev (m):	**0.0126**
Mean % error:	**0.437**

Summary statistics (1m):

Metric	Value
Mean displacement (m):	**0.987**
Std dev (m):	**0.0245**
Mean % error:	**1.01**



(The rest of these plots are in the zip file; they look mostly consistent)

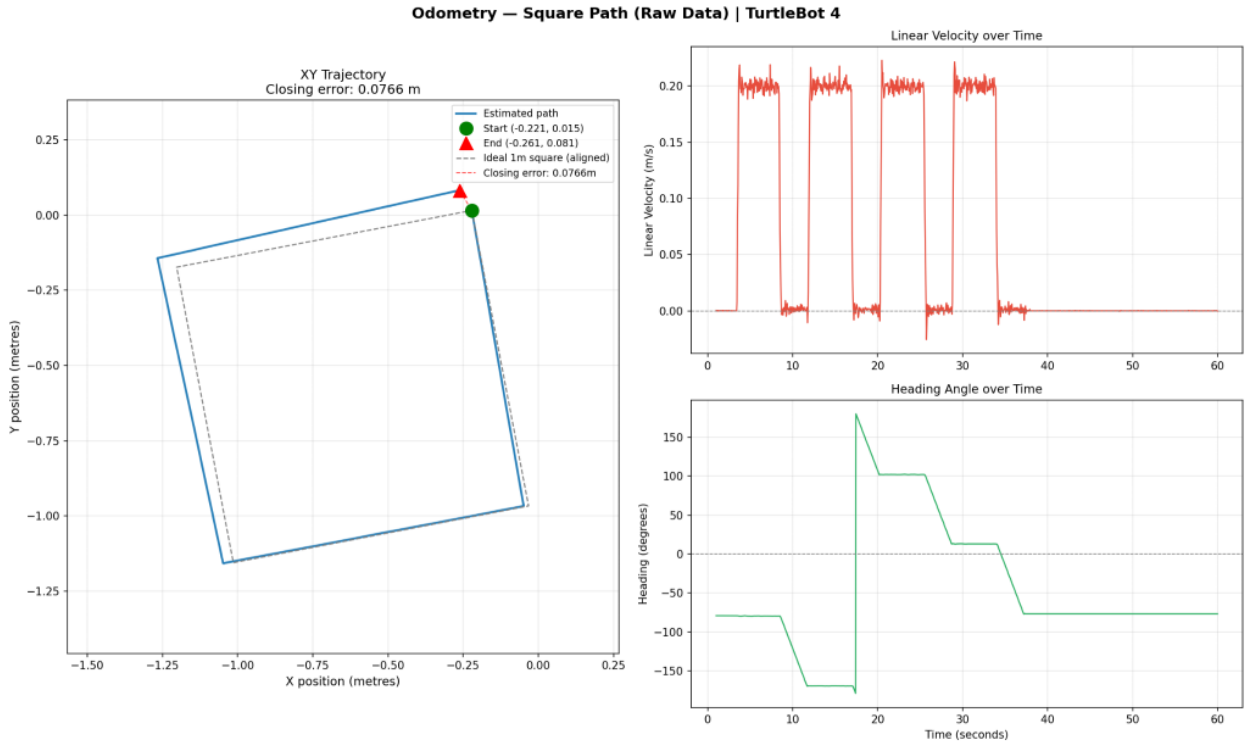
Test A2; Closed-Loop Drift

Extend your test node to drive a square and stop. The robot should end up back at the start. The odometry closing error, the distance between the reported final position and the origin, is your drift metric.

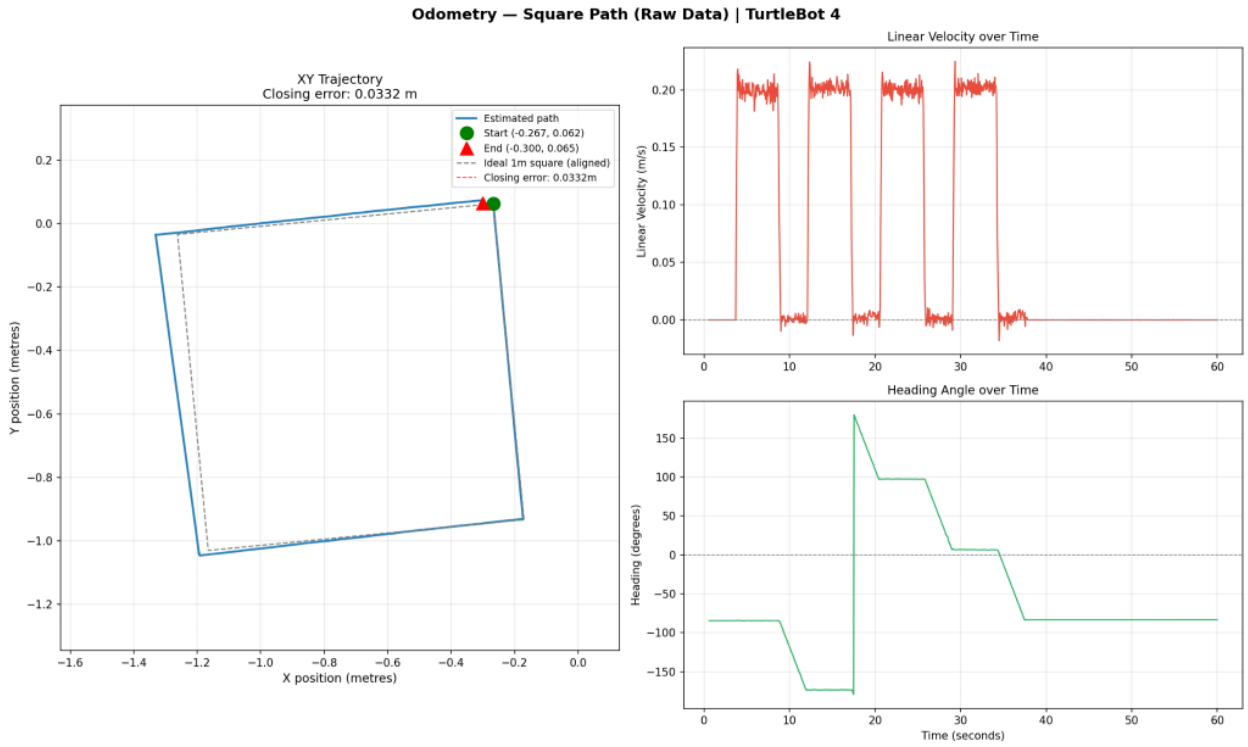
Pass/Fail criterion: physical closing error < 0.2m
Forward speed: 0.2m/s Turn:0.5 rad/s
1m x1m square

Trial	Odom closing error (m)	Physical closing error (m)	Final heading error (deg)	Pass / Fail
1	0.0766	0.09	2.45	Pass
2	0.0332	0.01	1.39	Pass
3	0.0224	0.02	1.75	Pass
4	0.0495	0.05	2.44	Pass
5	0.0517	0.03	1.82	Pass

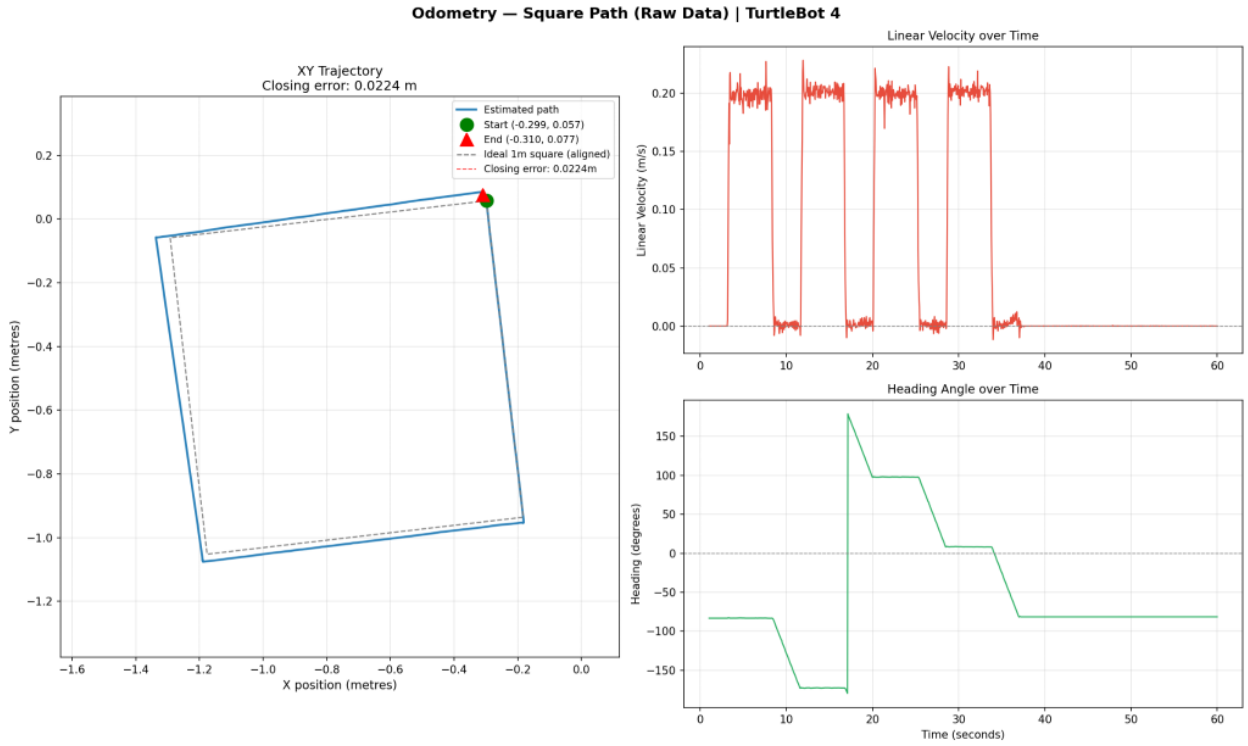
Mean physical closing error ≈ 0.04 m
Std dev ≈ 0.03 m (Physical closing error)
Mean odometry closing error ≈ 0.047 m
Mean heading error ≈ 1.97°



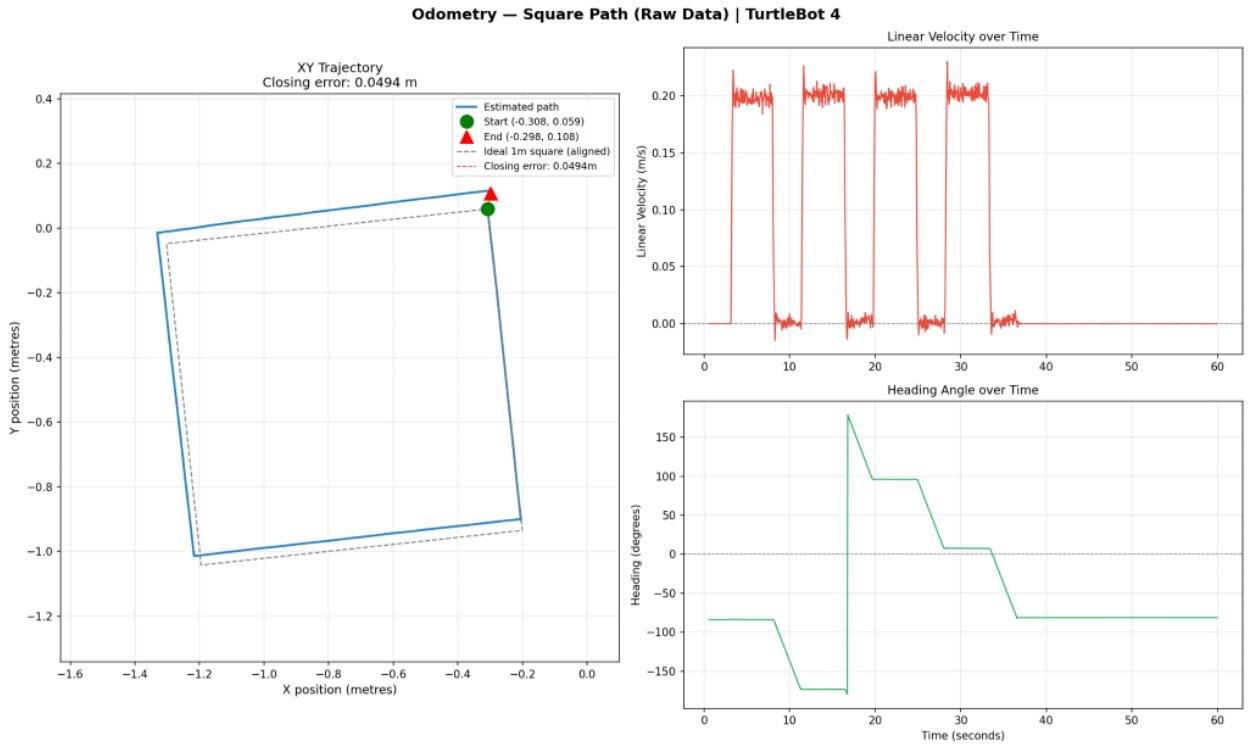
Samples: 1181 | Duration: 60.0s



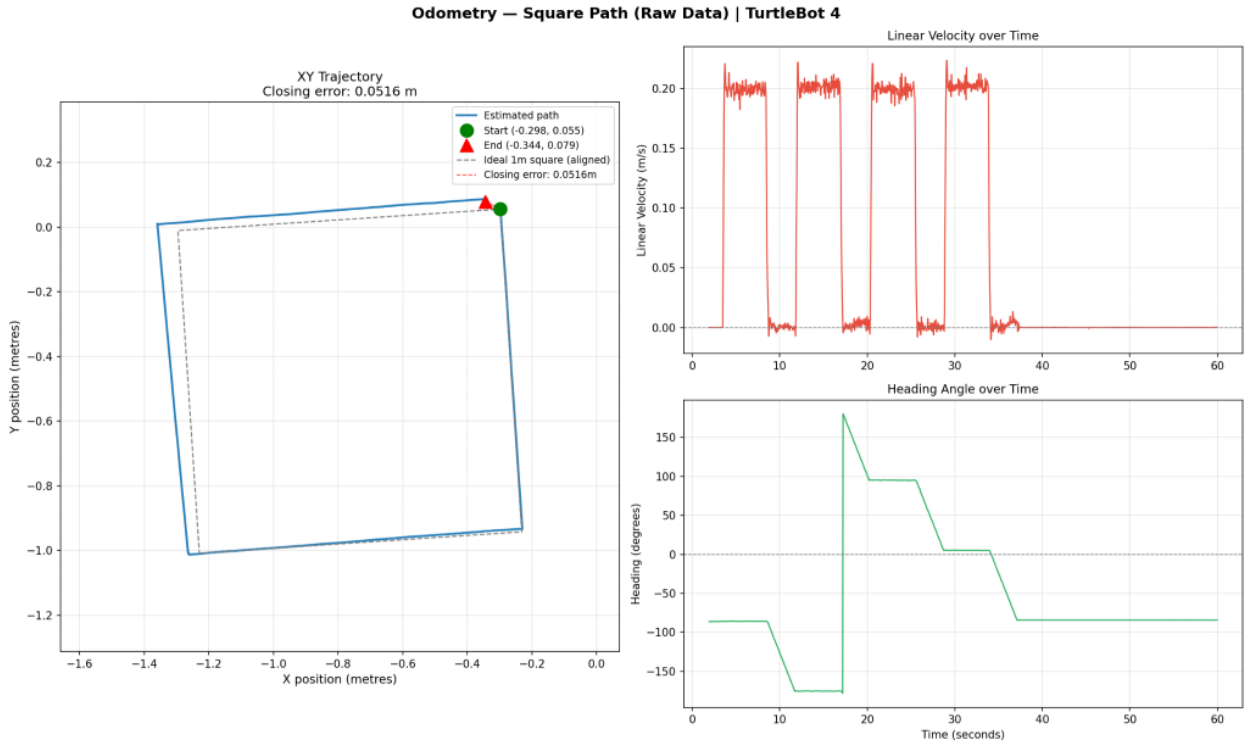
Samples: 1188 | Duration: 60.0s



Samples: 1178 | Duration: 60.0s



Samples: 1188 | Duration: 60.0s



Samples: 1162 | Duration: 60.0s

Investigation B; 2D LiDAR

The TurtleBot 4 uses an RPLIDAR-A1 sensor that produces a 360-degree ring of distance measurements. Your investigation must characterise range accuracy at known distances and measurement consistency.

Test B1; Range Accuracy at Known Distances

Pass/Fail criterion: % error < 2%

ground_truth_m	n_samples	mean_m	std_m	error_m	error_%	min_reading	max_reading
1	50	1.0269	0.0002	0.0269	2.69	1.0265	1.0273
1	50	1.0268	0.0002	0.0268	2.68	1.0264	1.0273
1	50	1.0267	0.0002	0.0267	2.67	1.0264	1.0273
1	50	1.0266	0.0002	0.0266	2.66	1.0262	1.0271
1	50	1.0267	0.0002	0.0267	2.67	1.0264	1.0273
2	50	2.0701	0.0004	0.0701	3.51	2.0691	2.0709
2	50	2.0698	0.0005	0.0698	3.49	2.0687	2.0705
2	50	2.0698	0.0004	0.0698	3.49	2.0691	2.0705
2	50	2.0696	0.0004	0.0696	3.48	2.0687	2.0705
2	50	2.0699	0.0004	0.0699	3.5	2.0691	2.0709
3	50	3.0753	0.0012	0.0753	2.51	3.0727	3.078
3	50	3.0747	0.0013	0.0747	2.49	3.0727	3.0777
3	50	3.0749	0.0014	0.0749	2.5	3.0724	3.0778
3	50	3.0755	0.0014	0.0755	2.52	3.0728	3.0789
3	50	3.0757	0.0014	0.0757	2.52	3.0727	3.0785

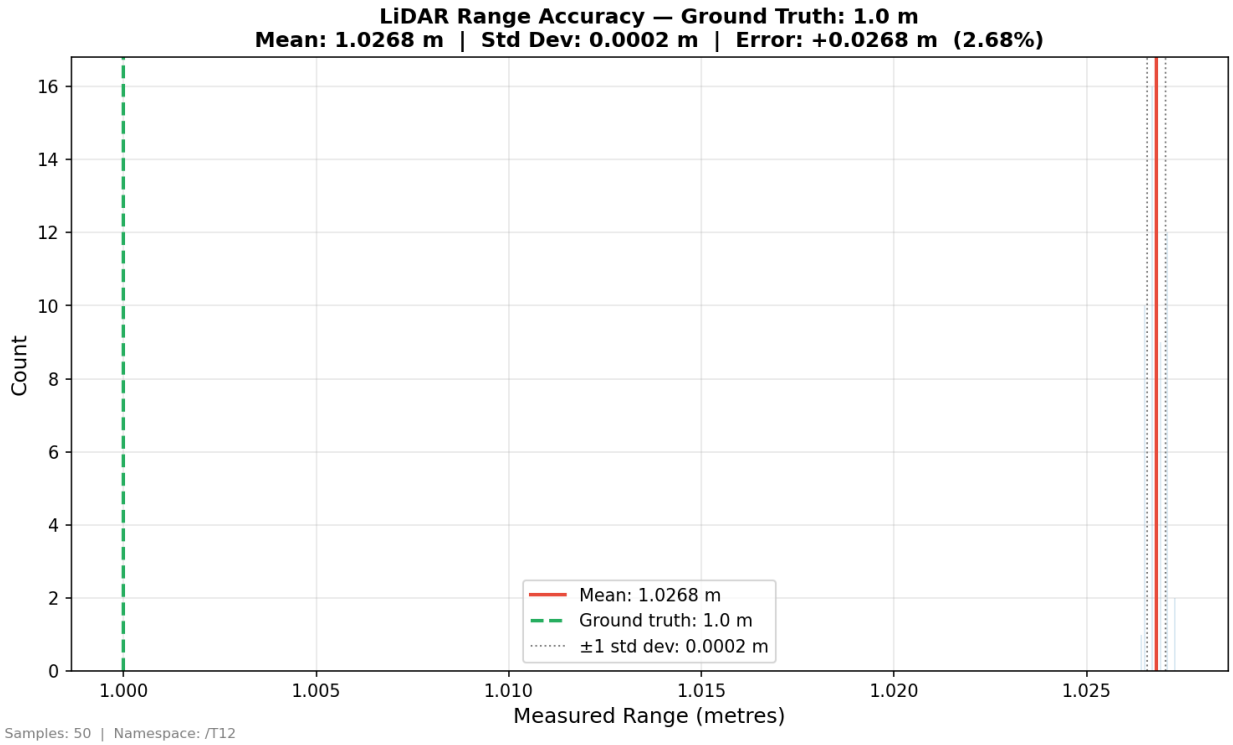
Summary Statistics 1m:
Mean measured distance: ≈ 1.0267 m
Mean error: ≈ 0.0267 m
Mean % error: ≈ 2.67%
Std deviation (avg): ≈ 0.0002 m

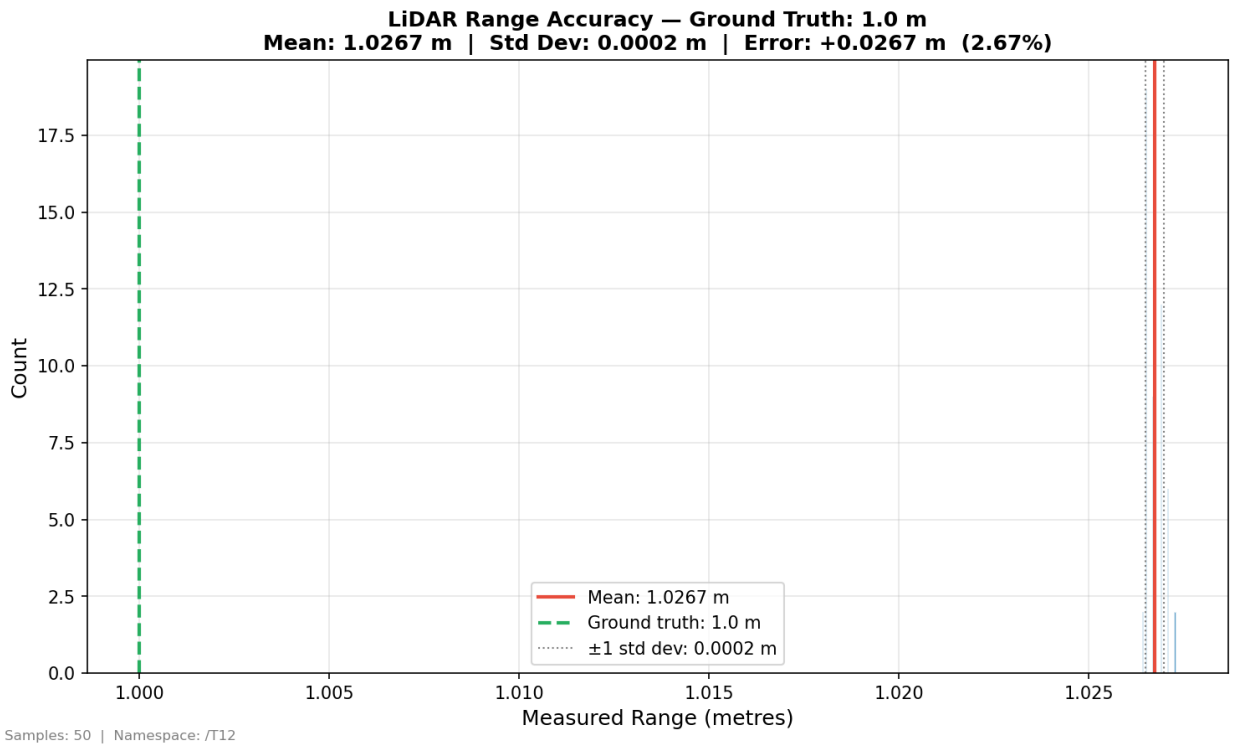
Summary Statistics 2m:
Mean measured distance: ≈ 2.0698 m
Mean error: ≈ 0.0698 m
Mean % error: ≈ 3.49%
Std deviation (avg): ≈ 0.0004m

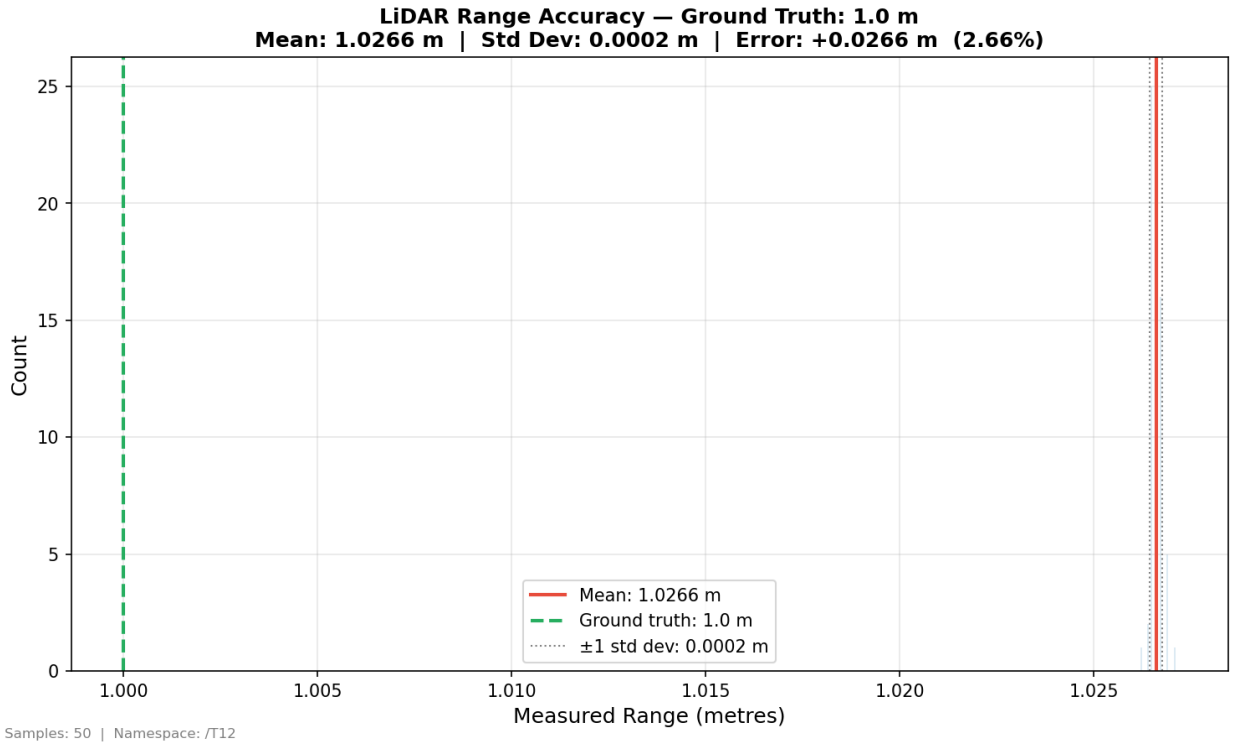
Summary Statistics 3m:
Mean measured distance: ≈ 3.0752 m
Mean error: ≈ 0.0752 m
Mean % error: ≈ 2.51%
Std deviation (avg): ≈ 0.0013 m

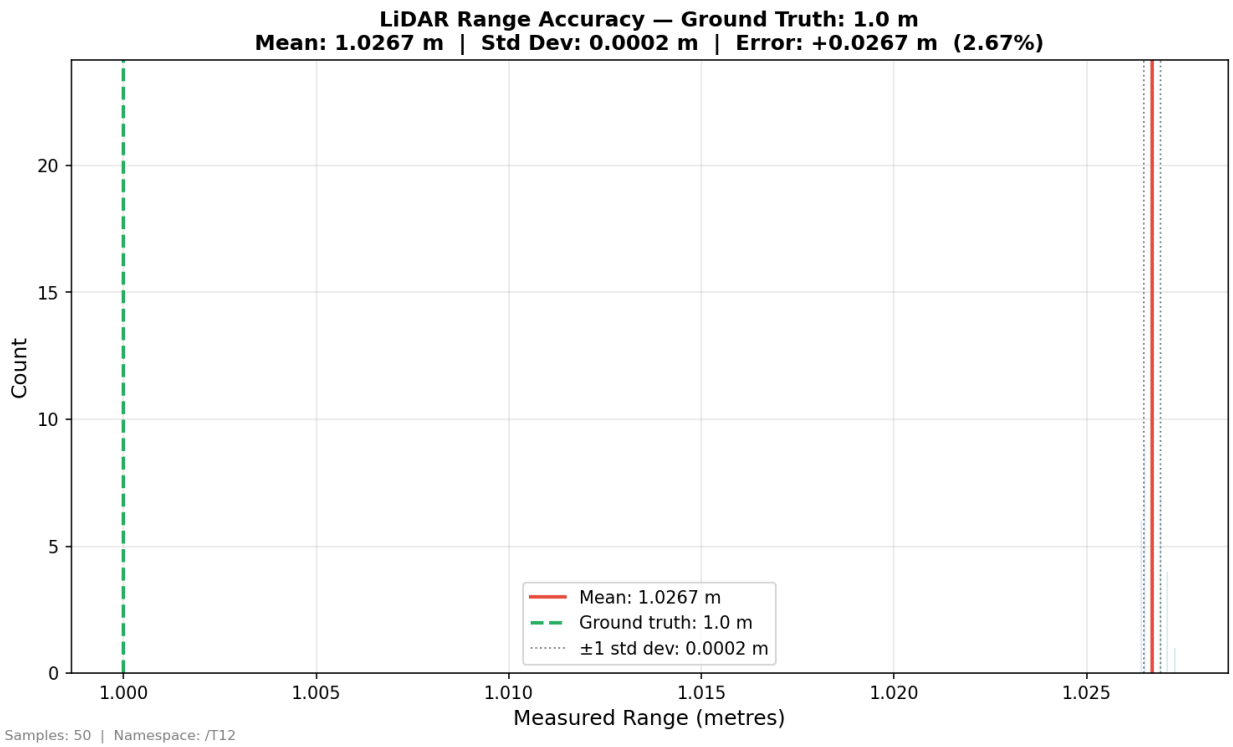
Histogram Notes:
The distributions are offset from the ground truth value, indicating a positive systematic bias in the measurements due to

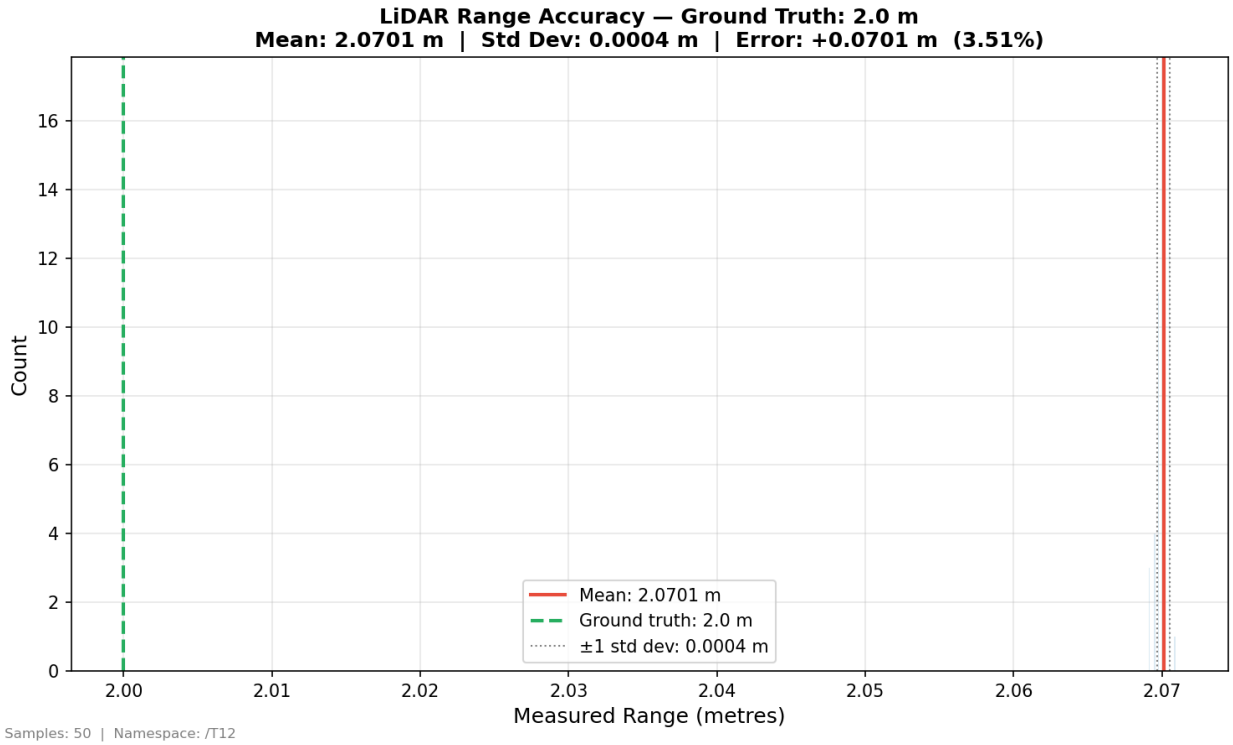
- Difficulty in aligning the robot precisely with floor markers
- uncertainty in positioning the LiDAR sensor origin relative to the measured distance
- misalignments between the robot and the wall

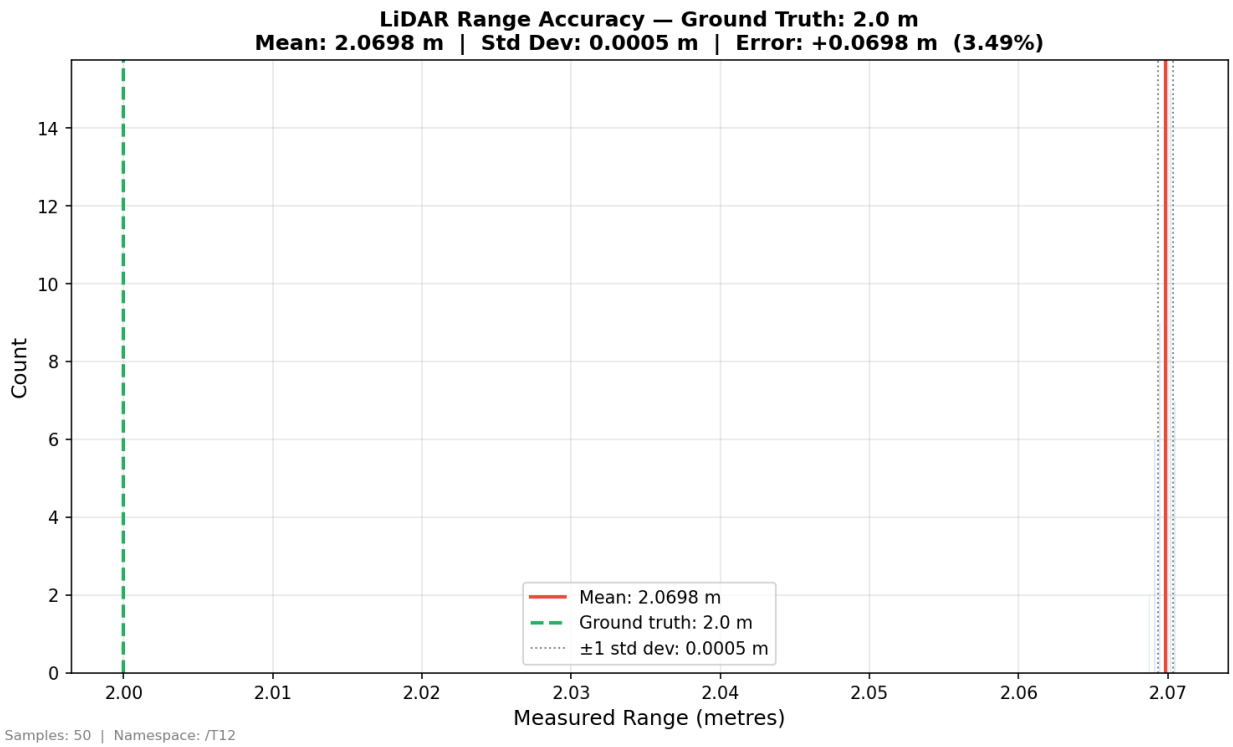

LiDAR Range Accuracy — Ground Truth: 1.0 m
Mean: 1.0268 m | Std Dev: 0.0002 m | Error: +0.0268 m (2.68%)
Count
Measured Range (metres)
Mean: 1.0268 m
Ground truth: 1.0 m
±1 std dev: 0.0002 m
Samples: 50 | Namespace: /T12

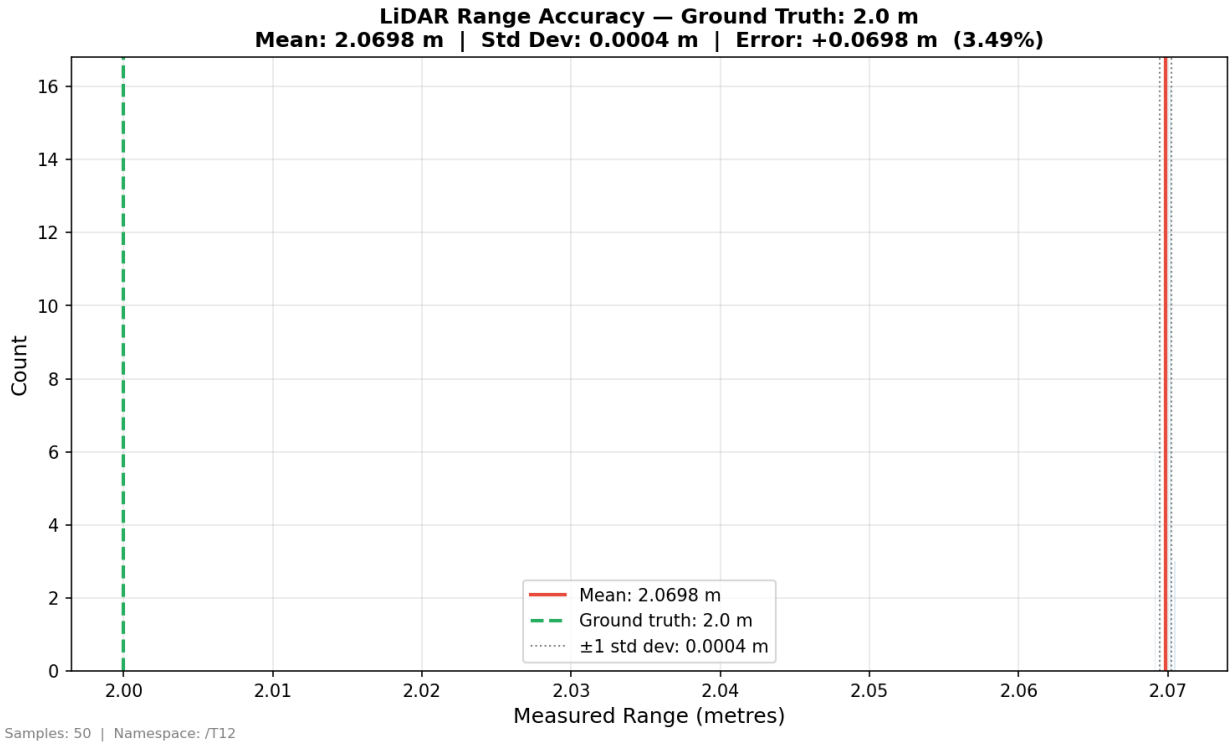

LiDAR Range Accuracy — Ground Truth: 1.0 m
Mean: 1.0267 m | Std Dev: 0.0002 m | Error: +0.0267 m (2.67%)
Count
Measured Range (metres)
Mean: 1.0267 m
Ground truth: 1.0 m
±1 std dev: 0.0002 m
Samples: 50 | Namespace: /T12

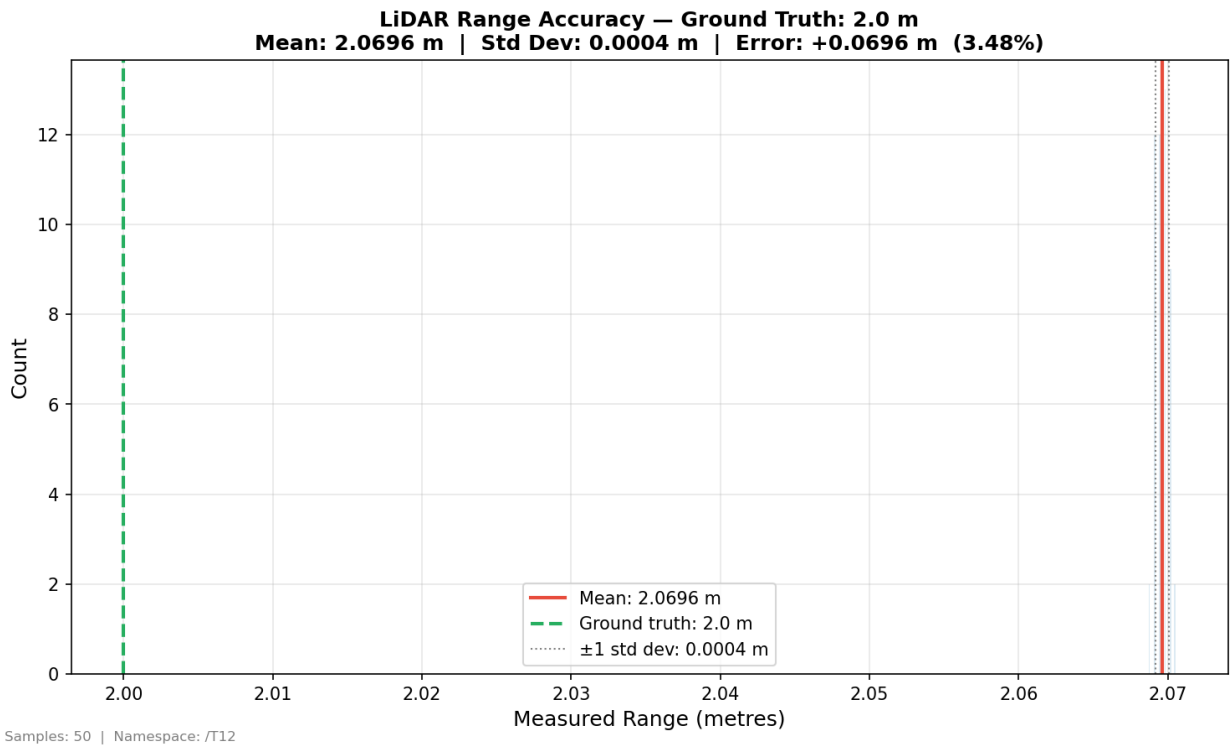

LiDAR Range Accuracy — Ground Truth: 1.0 m
Mean: 1.0266 m | Std Dev: 0.0002 m | Error: +0.0266 m (2.66%)
Count
Measured Range (metres)
Mean: 1.0266 m
Ground truth: 1.0 m
±1 std dev: 0.0002 m
Samples: 50 | Namespace: /T12

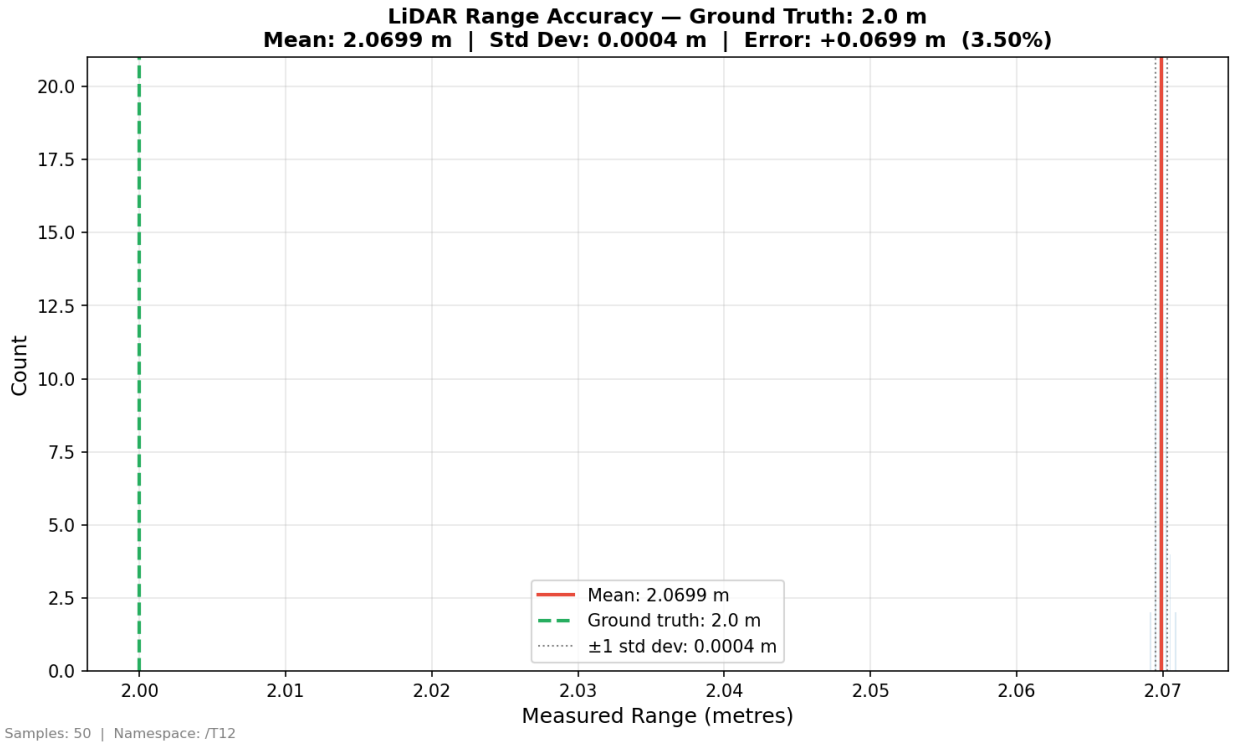

LiDAR Range Accuracy — Ground Truth: 1.0 m
Mean: 1.0267 m | Std Dev: 0.0002 m | Error: +0.0267 m (2.67%)
Count
Measured Range (metres)
Mean: 1.0267 m
Ground truth: 1.0 m
±1 std dev: 0.0002 m
Samples: 50 | Namespace: /T12

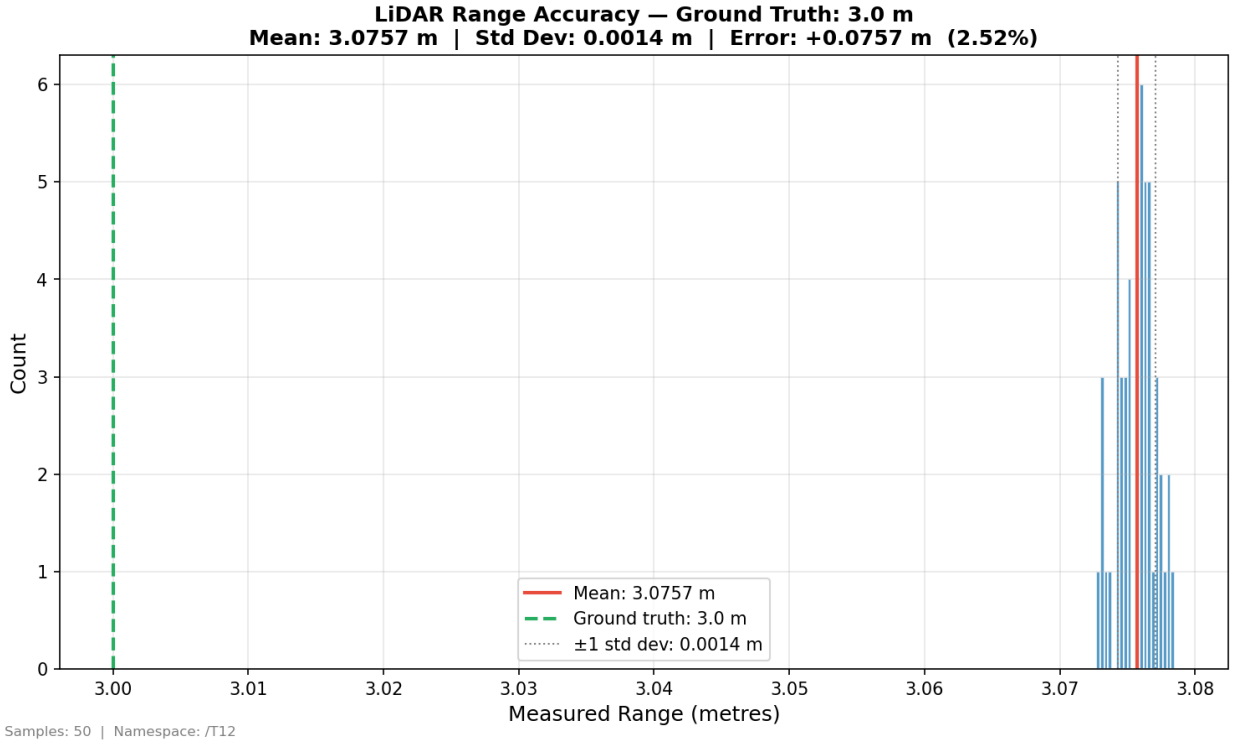

LiDAR Range Accuracy — Ground Truth: 2.0 m
Mean: 2.0701 m | Std Dev: 0.0004 m | Error: +0.0701 m (3.51%)
Count
Measured Range (metres)
Mean: 2.0701 m
Ground truth: 2.0 m
±1 std dev: 0.0004 m
Samples: 50 | Namespace: /T12

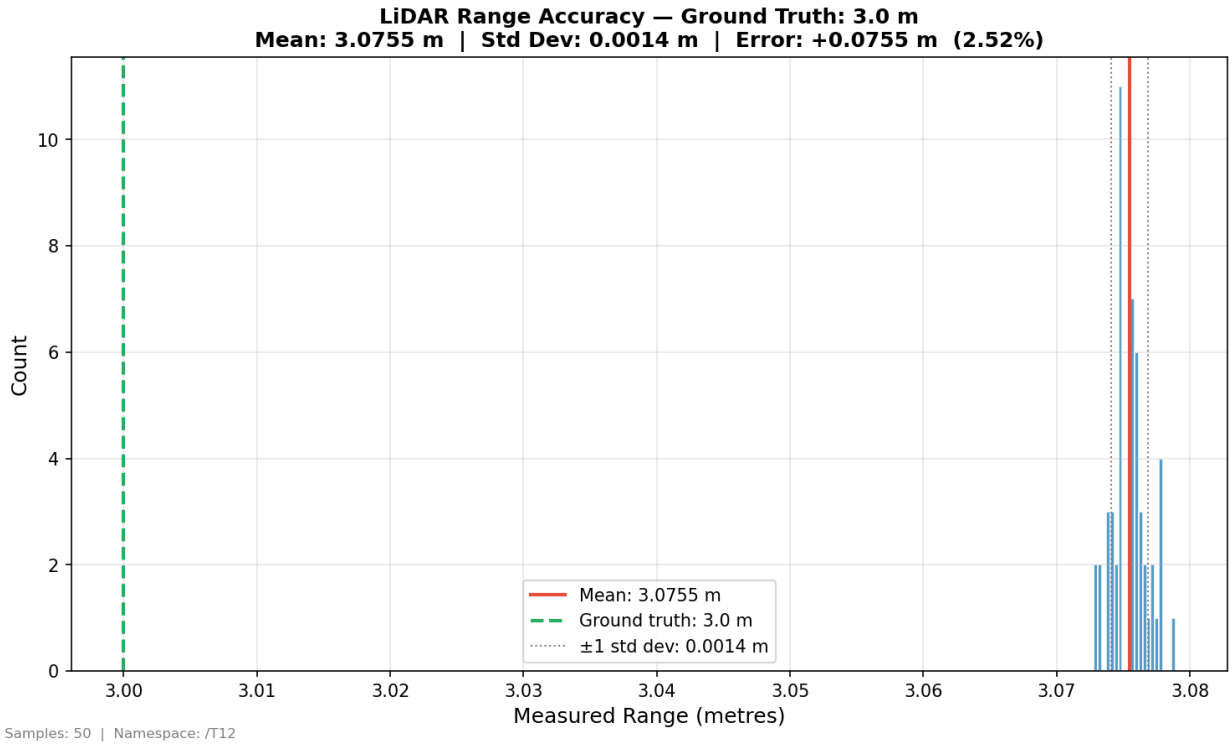

LiDAR Range Accuracy — Ground Truth: 2.0 m
Mean: 2.0698 m | Std Dev: 0.0005 m | Error: +0.0698 m (3.49%)
Count
Measured Range (metres)
Mean: 2.0698 m
Ground truth: 2.0 m
±1 std dev: 0.0005 m
Samples: 50 | Namespace: /T12

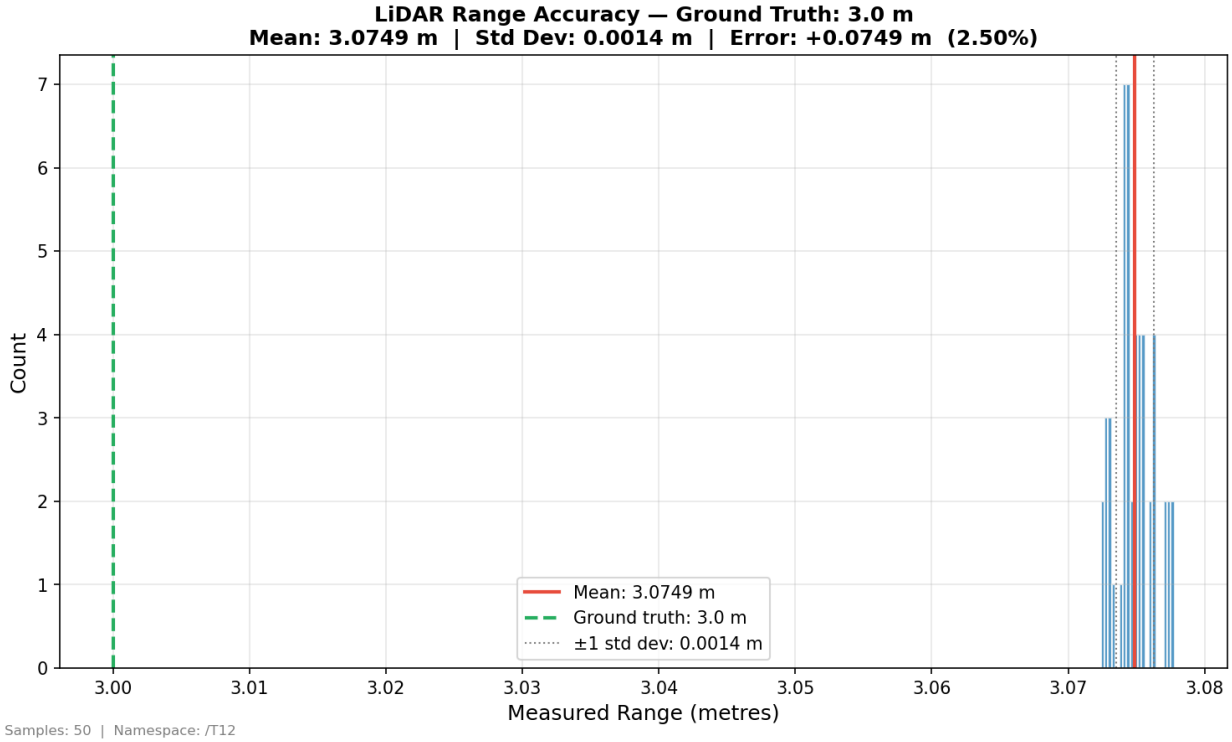

LiDAR Range Accuracy — Ground Truth: 2.0 m
Mean: 2.0698 m | Std Dev: 0.0004 m | Error: +0.0698 m (3.49%)
Count
Measured Range (metres)
Mean: 2.0698 m
Ground truth: 2.0 m
±1 std dev: 0.0004 m
Samples: 50 | Namespace: /T12

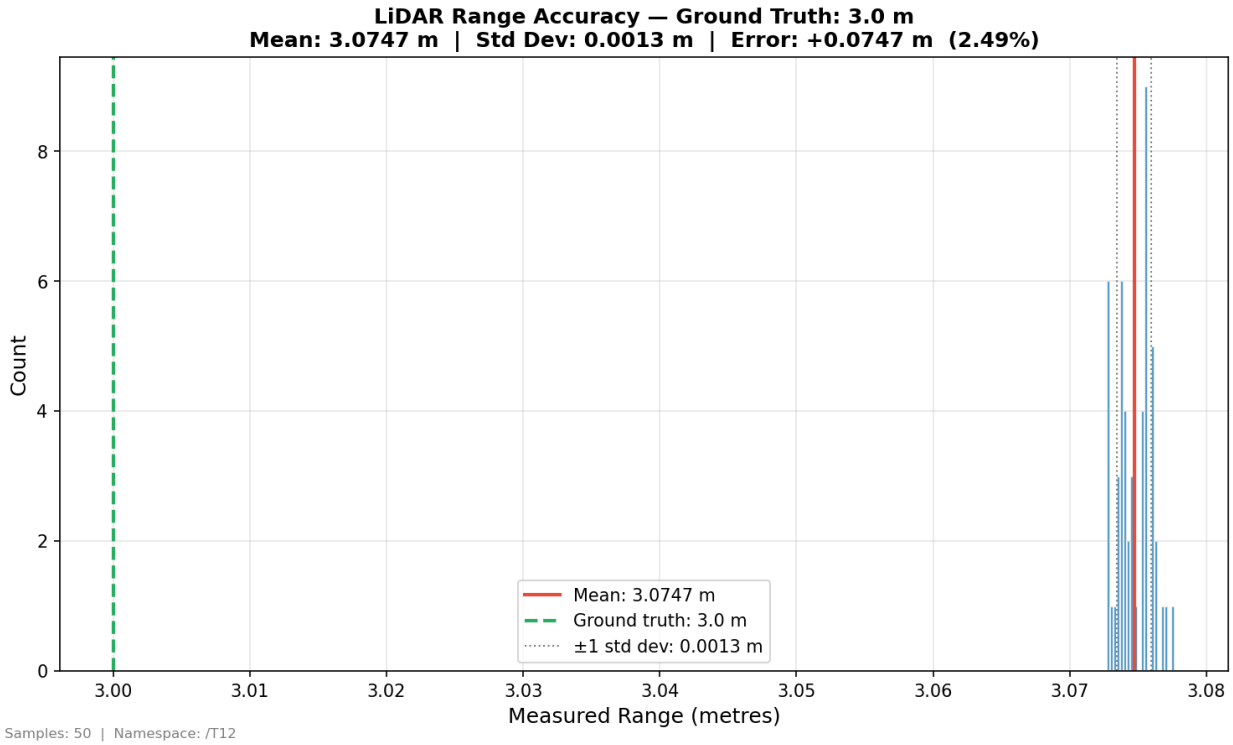

LiDAR Range Accuracy — Ground Truth: 2.0 m
Mean: 2.0696 m | Std Dev: 0.0004 m | Error: +0.0696 m (3.48%)
Count
Measured Range (metres)
Mean: 2.0696 m
Ground truth: 2.0 m
±1 std dev: 0.0004 m
Samples: 50 | Namespace: /T12

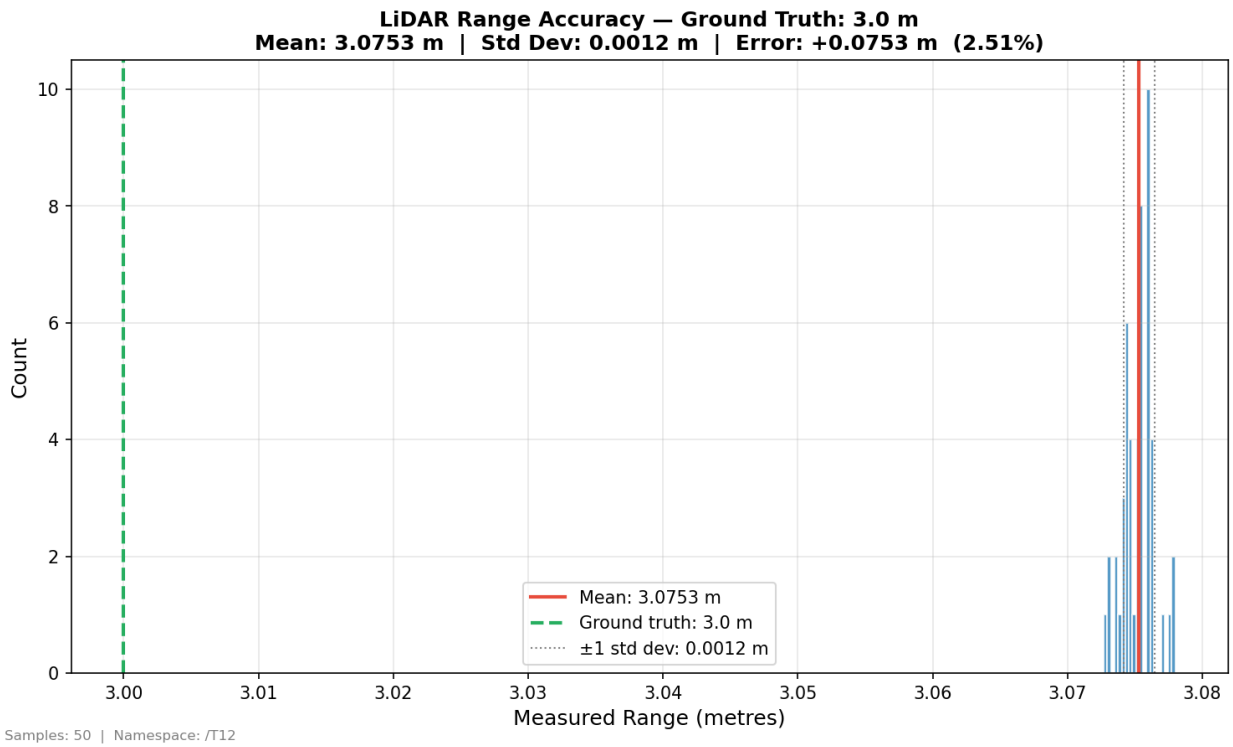

LiDAR Range Accuracy — Ground Truth: 2.0 m
Mean: 2.0699 m | Std Dev: 0.0004 m | Error: +0.0699 m (3.50%)
Count
Measured Range (metres)
Mean: 2.0699 m
Ground truth: 2.0 m
±1 std dev: 0.0004 m
Samples: 50 | Namespace: /T12


LiDAR Range Accuracy — Ground Truth: 3.0 m
Mean: 3.0757 m | Std Dev: 0.0014 m | Error: +0.0757 m (2.52%)
Count
Measured Range (metres)
Mean: 3.0757 m
Ground truth: 3.0 m
±1 std dev: 0.0014 m
Samples: 50 | Namespace: /T12


LiDAR Range Accuracy — Ground Truth: 3.0 m
Mean: 3.0755 m | Std Dev: 0.0014 m | Error: +0.0755 m (2.52%)
Count
Measured Range (metres)
Mean: 3.0755 m
Ground truth: 3.0 m
±1 std dev: 0.0014 m
Samples: 50 | Namespace: /T12

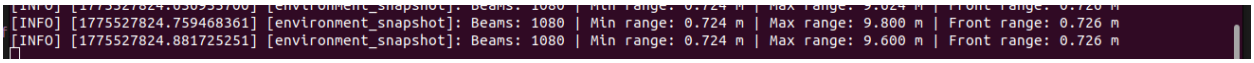

LiDAR Range Accuracy — Ground Truth: 3.0 m
Mean: 3.0749 m | Std Dev: 0.0014 m | Error: +0.0749 m (2.50%)
Count
Measured Range (metres)
Mean: 3.0749 m
Ground truth: 3.0 m
±1 std dev: 0.0014 m
Samples: 50 | Namespace: /T12

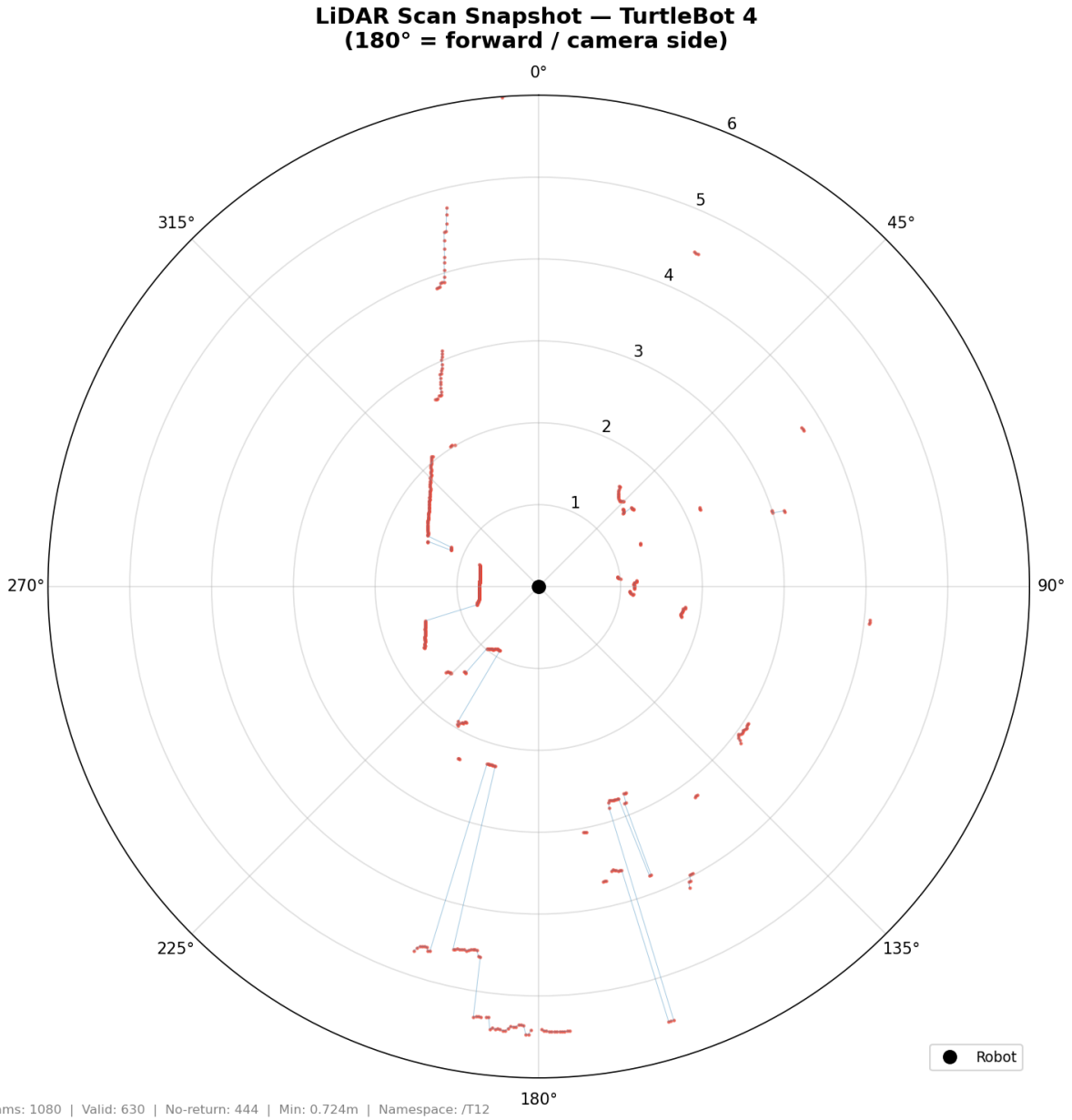

LiDAR Range Accuracy — Ground Truth: 3.0 m
Mean: 3.0747 m | Std Dev: 0.0013 m | Error: +0.0747 m (2.49%)
Count
Measured Range (metres)
Mean: 3.0747 m
Ground truth: 3.0 m
±1 std dev: 0.0013 m
Samples: 50 | Namespace: /T12


LiDAR Range Accuracy — Ground Truth: 3.0 m
Mean: 3.0753 m | Std Dev: 0.0012 m | Error: +0.0753 m (2.51%)
Count
Measured Range (metres)
Mean: 3.0753 m
Ground truth: 3.0 m
±1 std dev: 0.0012 m
Samples: 50 | Namespace: /T12

Test B2; Environment Snapshot

Observation	Value	Unit	Notes / Comments
Total beams per scan	1080	beams	Should be ~360 for RPLIDAR-A1 however we observed higher because the LiDAR driver outputs higher-resolution/interpolated scan data.
Minimum valid range observed	.724	m	Closest obstacle detected in lab
Maximum valid range observed	9.6	m	Open space in lab
Range straight ahead	.726	m	Pointing forward; facing wall or open area
Any anomalies noted	None	—	e.g., inf values, gaps, or noise




LiDAR Scan Snapshot — TurtleBot 4
(180° = forward / camera side)
0°
45°
90°
135°
180°
225°
270°
315°
1
2
3
4
5
6
Robot
Beams: 1080 | Valid: 630 | No-return: 444 | Min: 0.724m | Namespace: /T12

Investigation C; RGB Camera (OAK-D)

The TurtleBot 4 Standard is equipped with an OAK-D camera that captures a colour RGB image. Your investigation uses this camera to detect a red object in the scene using HSV colour thresholding; a fundamental technique used in robot vision for object detection and tracking.

Test C1; Live View and Observations

Verify the camera feed is working by opening a live view window. Point the camera at a variety of objects and surfaces and note what the camera can and cannot see clearly.

Camera feed works, one thing to note is that when it got too close to a glass door it looked like the glass wasn't even there.

Test C2; Red Object Detection

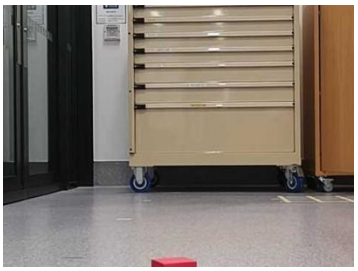
Place a red object in front of the camera at different distances. Run the detect mode to see the live detection overlay and observe how the pixel count changes with distance. The script saves a snapshot and CSV entry each time the object is detected above the threshold.

Day:

Distance (m)	Surface / Object	Red pixels detected	Object visible? (Y/N)	Notes
1.0	Cube	2870	Y	Had to lift the cube to see it Sleeve was used to cover hand as red pixels were being detected from hand. In the image below 6299 pixels were detected.

2.0	Cube	971	Y	
2.5	Cube	584	Y	
5	Cube	180	N	
4	Cube	350	N	

Night:

Distance (m)	Surface / Object	Red pixels detected	Object visible? (Y/N)	Notes
1.0	Cube	2803	Y	Had to lift the cube to see it  Sleeve was used to cover hand as red pixels were being detected from hand.

2.0	Cube	1031	Y	
3.0	Cube	1058	Y	
4.0	Cube	1001	Y	
5.0	Cube	623	Y	